Problem 08: Optimized Path Country

Time limit: 1 seconds

The country of Zelda has an extensive motorway system and has implemented a smart traffic system on its network. As soon as a driver enters a road network, an automated system demands the driver's destination to work out the exit toll gate and compute the optimal path. The automated system scans the car's tag and generates a coded sequence giving the optimal path and asks the driver to set his/her mileage to zero. In addition, the receipt also contains a token number which will be useful at the exit gate.

At the exit gate, the smart city has implemented a clever solution. If the driver has followed the optimal path, the car's tag will contain the total optimal distance covered as well as a number generated from the token that the car received at entrance. This is computed by adding the highest three prime numbers that are lower than the token. The toll is subsidized if the driver followed the optimal path, and a penalty is added if he/she took a longer route.

You want to take a romantic long drive on the motorway network but still pay a reduced toll. To do that, you need to enter the distance of the optimal path the code number. You are not to use any loops in your travel. Your point of entry is always denoted as town 1.

Input

In the first line has the road towns/intersection $N\ (1 \le N \le 100)$, followed by a number R corresponding to the number of roads in the network $R\ (1 \le R \le N \times N)$, your desired exit point $E\ (1 \le E \le 100)$, and the token assigned to you $T\ (1 \le T \le 32000)$. The subsequent R lines contain the distances $d\ (1 \le d \le 1000)$. corresponding to the road distances from towns u to $v\ (1 \le u,v \le 100)$.

Output

Your output should have as many lines as the number of test cases. Each line indicates the distance of your chosen path followed by the generated value from the token. In case there is no such path to your destination, the distance should be 2147483647.

Sample Input	Sample Output
4 4 3 10 1 2 2 1 3 3 2 3 4 3 4 1	3 15
4 2 3 15 1 2 2 3 4 1	2147483647 31